METRIC
DOD-M-24672(SH)
27 October 1986
SUPERSEDING
(See 6.5)

MILITARY SPECIFICATION

MOLDS, AND MOLD HEATING PLATEN FOR MOLDING ELECTRICAL CONNECTORS AND CABLES (METRIC)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers split cavity molds and a mold heating platen used for molding watertight and pressure-proof electrical connectors and for splicing cables for Naval use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-A-601 - Aluminum Alloy Sand Castings.
QQ-B-637 - Brass, Naval: Rod, Wire, Shapes, Forgings, and Flat Products with Finished Edges (Bar, Flat Wire, and Strip).
PPP-B-636 - Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-E-917	- Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-S-8559	- Steel Bars, Carbon, AISI (1095), Aircraft Quality.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5	- Plates, Identification.
MIL-M-24041	- Molding and Potting Compound, Chemically Cured, Polyurethane.
MIL-P-24074	- Polytetrafluoroethylene Parts and Coatings and Polyamide Parts.
MIL-C-24231	- Connectors, Plugs, Receptacles, Adapters, Hull Inserts, and Hull Insert Plugs, Pressure-Proof, General Specification For.

(See supplement 1 for list of associated specification sheets.)

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.
FED-STD-H28/2	- Screw - Thread Standards for Federal Services, Section 2, Unified Inch Screw Threads-UN and UNR Thread Forms.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MS28775	- Packing, Preformed, Hydraulic, +275°F ("O" Ring).
MS122081	- Insert, CRES Helical Coil Coarse Thread, 1 DIA Nominal Length.
MS122082	- Insert, CRES Helical Coil Coarse Thread, 1 DIA Nominal Length.
MS122121	- Insert, CRES Helical Coil Coarse Thread, 1-1/2 DIA Nominal Length.
MS122122	- Insert, CRES Helical Coil Coarse Thread, 1-1/2 DIA Nominal Length.

2.1.2 Other Government drawings and publication. The following other Government drawings and publication form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
NAVSHIPS
815-1197170 - Receptacle, Plug, Symbol 1170-1176.
815-1197172 - Receptacle Connector, Symbol 1229, 1230.
815-1197174 - Receptacle, Plug, Symbol 1177-1177.2.
815-1197176 - Receptacle Plug Connector, Symbol 1178.
815-1197304 - Plug H.S. RG-57 A/U Cable, Symbol 1119.
815-1197306 - Molding, Wiring Installation Procedures.
815-1197328 - Plug Assembly Limit Switch, Symbol 1223.
815-1197332 - Plug Assembly, Resistance Temperature, Symbol 1224.
815-1197335 - Plug, Ungrounded RG-17A/U, Symbol 1225.
815-1197343 - Molding Wiring Procedures WT Splice.
815-1197348 - Receptacle Plug Connector, Symbol 1240-1243.
815-1197349 - Receptacle Plug Connector, Symbol 1248, 1249.
815-1197352 - Receptacle Plug Connector, Symbol 1252, 1253.
815-1197353 - Receptacle, Plug Connector, Symbol 1254, 1255.
815-1197354 - Receptacle Plug Connector, Symbol 1250, 1251.
815-1197376 - Receptacle, Plug Connector, Symbol 1244.
815-1197377 - Receptacle Plug Connector, Symbol 1245-1245.2.
815-1197381 - Molding, Wiring Procedures for Connectors Electrical.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
0962-LP-022-2010 - Molding and Inspection Procedures for Fabricating Connector Plugs for Submarine Outboard Cables.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
B46.1 - Surface Texture, Surface Roughness, Waviness, and Lay. (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A 29 - Standard Specification for Steel Bars Carbon and Alloy, Hot-Wrought and Cold Finished, General Requirements for. (DoD adopted)
A 108 - Standard Specification for Steel Bars, Carbon, Cold-Finished, Standard Quality. (DoD adopted)

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B 211 - Standard Specification for Aluminum-Alloy Bar, Rod, and Wire. (DoD adopted)
D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.3 Materials. Materials shall be as specified (see 3.1). When not specified, the contractor may select any material that shall satisfactorily perform the intended function in the equipment and shall comply with the requirements of this specification.

3.3.1 Metals. Metals shall be of the corrosion-resistant type or shall be treated to resist corrosion in accordance with MIL-E-917.

3.3.2 Prohibited materials. Materials listed in MIL-E-917 with the following additions shall not be used:

(a) Asbestos
(b) Cadmium
(c) Mercury (see 3.3.2.1)

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3.3.2.1 Mercury. In addition to the mercury requirement of MIL-E-917, no parts shall have come in direct contact with metallic mercury, mercury compounds or mercury bearing instruments or devices employing only a single boundary of containment during manufacture or inspection. (A single boundary of containment is one which is not backed up by a second seal or barrier to prevent escape and consequent mercury contamination in the event of rupture of the primary seal or barrier.)

3.3.3 Recovered materials. Unless otherwise specified herein, all material and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4 Construction. Molds and the heating platen shall be of the construction and physical dimensions specified (see 3.1).

3.4.1 Surface finishes. Surface finishes of machined metal parts shall be as specified (see 3.1) and shall be in accordance with ANSI B46.1.

3.4.2 Drilling, countersinking, and tapping. Drilling, countersinking, and tapping metal parts shall be done before plating, coating, or finish is applied.

3.4.3 Threaded parts. Screw threads shall be in accordance with FED-STD-H28. Unless otherwise specified (see 3.1), threads shall be right hand, coarse-thread series, unified form, in accordance with class 2A or 2B of FED-STD-H28/2.

3.4.4 Welding and brazing. Welding and brazing practices shall be in accordance with commercial practices.

3.4.5 Polytetrafluoroethylene (TFE). Unless otherwise specified (see 3.1), a commercial grade TFE coating, where required, shall be applied by the impregnation treatment method to provide a uniform coating to those surfaces treated.

3.5 Performance requirements.

3.5.1 Molds (leakage). Molds shall exhibit no signs of leakage of molding compound when tested as specified in 4.6.2.

3.5.2 Heating platen.

3.5.2.1 Leakage current. Leakage current shall not exceed 5 milliamperes when the heating platen is tested as specified in 4.6.3.1.

3.5.2.2 Dielectric withstanding voltage. The electrical circuits shall exhibit no evidence of disruptive discharge, arcing, flashover, corona, or breakdown when the heating platen is tested as specified in 4.6.3.2.

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3.5.2.3 Insulation resistance. The insulation resistance of the heating platen heater contacts shall be 50 megohms or greater and the insulation resistance of the cable assembly shall be 200 megohms or greater when tested as specified in 4.6.3.3.

3.5.2.4 Operation. The thermostat control and thermometer readings shall remain within plus or minus 10 percent of each other when the heating platen is tested as specified in 4.6.3.4.

3.6 Identification plates. Identification plates shall be in accordance with MIL-P-15024 and MIL-P-15024/5, normal service. Marking information shall be as specified (see 3.1).

3.7 Workmanship. Parts shall be free of burrs, sharp edges, or any other damage or defect that could make them unsatisfactory for the purpose intended.

3.7.1 Threaded parts or devices. Screws, nuts, and bolts shall show no evidence of cross threading, mutilation, detrimental or hazardous burrs.

3.7.2 Castings. Castings shall be free from cold shuts, blow holes, or any imperfections that may affect strength of function in any way.

3.7.3 TFE coating. The TFE coating shall be free of cracks or other imperfections which would produce an uneven surface.

3.7.4 Cleaning. After fabrication or machining, parts and assembled equipment shall be cleaned of smudges, metal chips or any other foreign material which might detract from the intended operation, function or appearance of the equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in MIL-STD-202.

4.4 First article inspection. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production (see 6.3).

4.4.1 Sample. One sample of each type mold or heating platen shall be subjected to first article inspection.

4.4.2 Inspection routine. Sample units shall be subjected to the first article inspection specified in table I.

TABLE I. First article inspection.

Inspection	Requirement	Test method
All samples:		
Visual and dimensional examination	3.1, 3.3, 3.4, 3.6 and 3.7	4.6.1
Split-cavity molds (only):		
Leakage	3.5.1	4.6.2
Heating platen only:		
Leakage current	3.5.2.1	4.6.3.1
Dielectric withstanding voltage	3.5.2.2	4.6.3.2
Insulation resistance	3.5.2.3	4.6.3.3
Operation	3.5.2.4	4.6.3.4

4.4.3 Failures. One or more failures shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be as specified in table II.

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TABLE II. Quality conformance inspection.

Inspection	Requirement	Test method
All samples: Visual and dimensional examination	3.1, 3.3, 3.4, 3.6 and 3.7	4.6.1
Heating platen only: Operation	3.5.2.4	4.6.3.4

4.5.1 Inspection lot. An inspection lot shall consist of molds or heating platens of the same basic design, produced under essentially the same conditions, and offered for inspection at one time.

4.5.2 Sampling plan. Statistical sampling and inspections shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be 1.5 percent defective.

4.6 Methods of inspection.

4.6.1 Visual and mechanical examination. Molds and the heating platens shall be visually examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, 3.6, and 3.7).

4.6.2 Split cavity molds (leakage). The split cavity molds shall be tested for leakage of molding compound using a modified version of the molding procedures specified in the applicable specification sheet (see 3.1). The following options and exceptions apply:

(a) At the option of the contractor, the molding compound for leakage tests may be polyurethane in accordance with type I or II of MIL-M-24041.
(b) Cable preparation and wiring may be omitted.
(c) A heating platen or oven may be used to cure the compound.
(d) Actual plug inserts/insulators may be used. If simulated inserts/insulators are used, they shall be of the same overall dimensions as the actual items.
(e) The cable shall be of the same size as that specified in the procedures.
(f) Acceptance criteria: There shall be no signs of mold compound leakage outside the mold.

4.6.3 Heating platen.

4.6.3.1 Leakage current. The leakage current shall be measured in accordance with the procedures for portable equipment of MIL-E-917.

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4.6.3.2 Dielectric withstanding voltage. The heating platen shall be subjected to the dielectric withstanding voltage test in accordance with method 301 of MIL-STD-202. The following details shall apply:

(a) Test voltage: 1225 volts.
(b) Nature of potential: alternating current (ac)
(c) Points of application: Between manually insulated points and between insulated points and ground (heating platen).
(d) Operating control: The thermostat switch shall be in the on position.

4.6.3.3 Insulation resistance. Insulation resistance shall be measured in accordance with method 302 of MIL-STD-202. The following details shall apply:

(a) Test condition: B
(b) Points of application: (1) With the cable assembly disconnected, between heater contacts and ground. (2) With the cable assembly connected, between the electrical conductors and between electrical conductors and ground. The thermostat shall be in the off position and measurements shall be taken at the ground power plug only.

4.6.3.4 Operation. The heating platen shall be operated at the low, medium and high ranges. The temperature shall be allowed to stabilize at each setting and the thermometer reading shall be compared to the setting on the calibrated dial (see 3.5.2.4).

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2.1).

5.1.1 Level A. Each heating platen shall be preserved in accordance with submethod IIb of MIL-P-116. The molds shall be individually preserved in accordance with method III of MIL-P-116.

5.1.2 Commerical. Heating platens and molds shall be individually preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level B, C or commercial, as specified (see 6.2.1).

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5.2.1 Level B. Items, preserved as specified in 5.1 shall be packed in fiberboard boxes conforming to PPP-B-636, class weather resistant with type, variety, grade and style selection at the contractor's option. Box closure and reinforcement shall conform to method V of the appendix to the box specification.

5.2.2 Level C. Items preserved as specified in 5.1 shall be packed in fiberboard boxes as specified in level B except that boxes shall be of the domestic class and box closure shall conform to method I of the appendix to the box specification.

5.2.3 Commercial. Items preserved as specified in 5.1 shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required (see 6.2.1), interior and exterior containers packed levels B and C shall be marked in accordance with MIL-STD-129. Commercial packs shall be marked in accordance with ASTM D 3951. Bar code markings shall be applied to all interior and exterior containers in accordance with MIL-STD-129.

6. *NOTES*

6.1 Intended use. The items covered by this specification are intended to be used for molding watertight and pressure-proof connectors and for splicing cables.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification and applicable specification sheet.
(b) When first article inspection is required (see 3.2).
(c) Preservation and packing, level or commercial requirements (see 5.1 and 5.2).
(d) Special marking required (see 5.3).

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6.5 Supersession data. The following NAVSHIPS drawings listed in table III have been superseded by this specification and the applicable specification sheets:

TABLE III. Cross index of drawings and specification sheets.

Drawing	Specification sheet
815-1197171	DOD-M-24672/6
815-1197173	DOD-M-24672/6
815-1197175	DOD-M-24672/6
815-1197251	DOD-M-24672/1, /3
815-1197252	DOD-M-24672/1
815-1197253	DOD-M-24672/1
815-1197256	DOD-M-24672/2
815-1197303	DOD-M-24672/6
815-1197330	DOD-M-24672/4
815-1197333	DOD-M-24672/10
815-1197334	DOD-M-24672/7
815-1197339	DOD-M-24672/11
815-1197355	DOD-M-24672/5
815-1197356	DOD-M-24672/5
815-1197357	DOD-M-24672/5
815-1197358	DOD-M-24672/5
815-1197359	DOD-M-24672/5
815-1197360	DOD-M-24672/9
815-1197378	DOD-M-24672/4
815-1197379	DOD-M-24672/8

6.6 Subject term (key word) listing.

Connectors
Electrical
Heating platen
Molds
Pressure-proof
Splicing cables
Split cavity

Preparing activity:
Navy - SH
(Project 5935-N263)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-M-24672(SH)	MOLDS, AND MOLD HEATING PLATEN FOR MODLING ELECTRICAL COMMECTORS AND CABLES (METRIC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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